
Designer Training

Course Overview

Description:

- This beginner training is for users that are responsible for designing, building and maintaining data models. The training will discuss the basics of data modeling and will teach users how to connect resources through mapping. Users will learn how to build out a model, publish it and then view it in Explorer. They will learn how to build out inference rules and then apply these rules to their data model.

Audience:

- New Stardog users that are consumers, modelers or builders of knowledge graphs.

Pre-requisites:

- User should have a Stardog Free Cloud account and endpoint
- Users should complete the Meet Stardog and Explorer training

Course Objectives

After completing this training, students will be able to:

- Understand basic data modeling and terminology
- Create mappings to project resources
- Create and map relationships
- Create and edit attributes and labels
- Publish a model
- Create and publish inference rules

Stardog Cloud

- Manage Knowledge Graph endpoints
- Launch applications
- Sign up for Stardog Cloud plans
- Access Resource Center for guided journeys and training resources
- Invite more users

The screenshot shows the Stardog Cloud portal interface. At the top, a dark header bar contains the text "Welcome to Stardog Cloud" and a link to the "Stardog Community". Below this, the main content area is titled "Stardog Training 2". It features several tiles: "DESIGNER" (A no-code, visual environment to model, map, and publish data for your Knowledge Graph), "EXPLORER" (An intuitive interface to visually browse and query your Knowledge Graph), and "STUDIO" (A complete IDE to program your Knowledge Graph). A "Metrics" section displays a table with the following data:

Metrics	
Edges	420,580
Nodes	40,903
Queries	17,559

Below the metrics, there are "Quick Actions" including "Invite users", "Manage endpoints", and "Manage Databricks connections". On the right side, there are links to "Knowledge Kits", "Docs", "Trainings", and "Community". A sidebar on the left contains a "PORTAL" header and a "Stardog Training 2" link. At the bottom left, there is a "Learn Stardog" link, and at the bottom right, there is a "Stardog" logo.

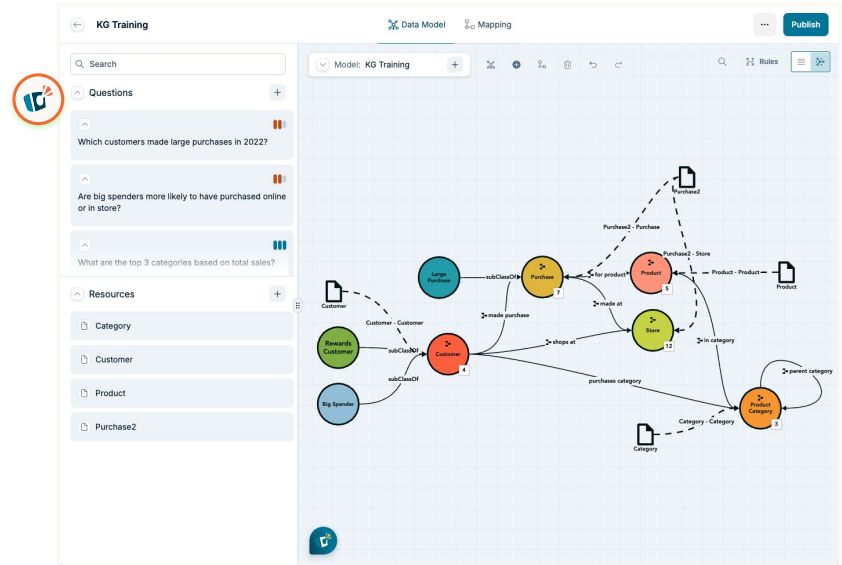
Stardog Designer

Stardog Designer

No-code, visual KG authoring

Features:

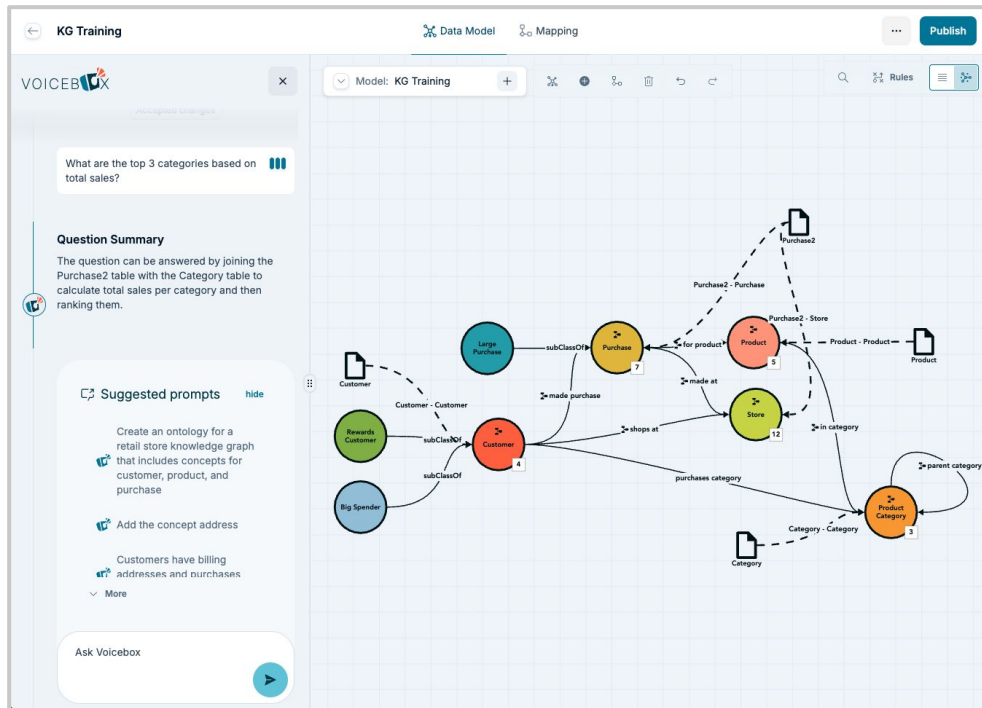
- Create & Edit data models
- Load existing models
- Define inference rules
- Knowledge Catalog powered mapping



Voicebox in Designer

Use Voicebox as an expert user that can help you:

- Create new concepts based on core concepts
- Extend and modify the data model
- Map project resources to the data model
- Available for Voicebox enabled endpoints
- Voicebox only features highlighted with 



Agenda

-
- 1 Data Modeling
 - 2 Project Resources & Mapping
 - 3 Relationships
 - 4 Publishing the Model
 - 5 Inference Rules
-

1

Data Modeling

- Data Modeling Basics
- Naming Conventions
- Terminology
- Building & Editing the Model

Getting Started with Your First Data Model

- Start with a few key concepts or your data
- Data models should be composable like Legos, reuse is key
- Naming
 - Data models encode the meaning of your data
 - Terminology should align with domain experts

Data Modeling Approaches

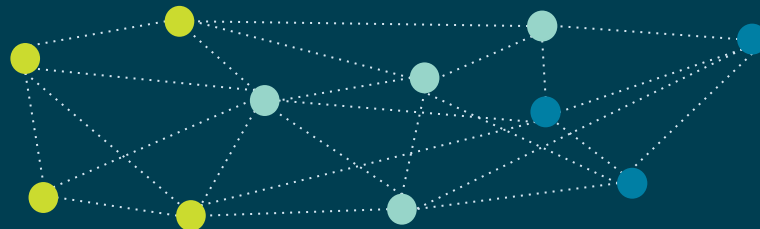
Top-Down



Bottom-Up



Enterprise
Knowledge
Graph



Retail Store

Customer

Product

Order

Inventory

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

Products

id	product	msrp	dept
8.79% distinct 80 records	99.83% distinct 580 records	4.66% distinct 579 records	
	Ring Light	568.66	8
	Wireless Charger	275.17	8
	Notebook	492.14	23
	Snowglobe	589.83	21

Ways to Start

- Create the data model by hand (**Top-Down**)
- Use the data (CSVs or SQL DBs) to inform the data model (**Bottom-Up**)
- Load the data model from another Designer project
- Import the data model from an existing Stardog database
- Load the data model from an Ontology editor (e.g. Protégé)
- Load an open source ontology

Naming conventions

Pick a convention for naming things - any convention - and stick with it.

Entity and Relationship Naming Conventions:

- **Entity Names:**
 - a. Use **singular form** for class names (e.g., **Customer**, **Product**, **Purchase**), as it represents a single instance of the entity.
- **Relationship Names:**
 - a. Use **camelCase**, with the verb in the **present tense**, or preposition followed by the destination class (e.g., **buysProduct** for the relationship between **Customer** and **Product**).
- **Attribute Names:**
 - a. Avoid verbs, and describe them in a **concise and descriptive** manner (e.g., **Age**, **Price**, **Name**).

*Avoid unnecessary prefixes or abbreviations to keep names intuitive.

Top Down Example

Use Case Overview

The use case for this course is based on a **Customer 360 analysis**.

The customer in this case is someone that buys products from the company's ecommerce site or in a brick and mortar store.

Here are some questions that users will ask of this use case:

- What do customers buy and how often?
- Do customers use loyalty accounts?
- What type of payment methods do they use?
- What are the buying patterns for customers based on location, age, gender, etc?
- What products are typically bought together?
- Are there any patterns related to when the customer buys in a store vs. online?

Identify Business Concepts

In Designer, the user will create the data model to support the use case. Before using Designer, think about the data model and start to identify business concepts.

Customers

Purchase

Loyalty Account

Payments

Store Location

Gender

Product

Year

Customer Location

Store Type

Create Ontological Statements

Use the business concepts to form ontological statements. For example, Customer buys products

- Customer made purchase
- Customer has a Loyalty Account
- Products are in a category
- Categories have a parent category
- Customer shops in a physical store
- Customer shops online
- Customer is female
- Purchase made with a visa
- Purchase made in 2022
- In store shoppers are customers who have an in store purchase

Noun & Verb Analysis

Identify the **nouns** and **verbs**:

- Customer **made** a Purchase
 - Customer **made** purchase with a Loyalty Account
 - Products **are** in a category
 - Categories **have** a parent category
 - Customer **shops** in a physical store
 - Customer **shops** online
 - Customer **is** female
 - Purchase **made** with a visa
 - Purchase **made** in 2022
 - *In store shoppers* are Customers who have an *in store purchase*
-
- In Designer the nouns will turn into instances of **Classes**, or **Attributes values** (= literal values) and the verbs will become **Relationships**.

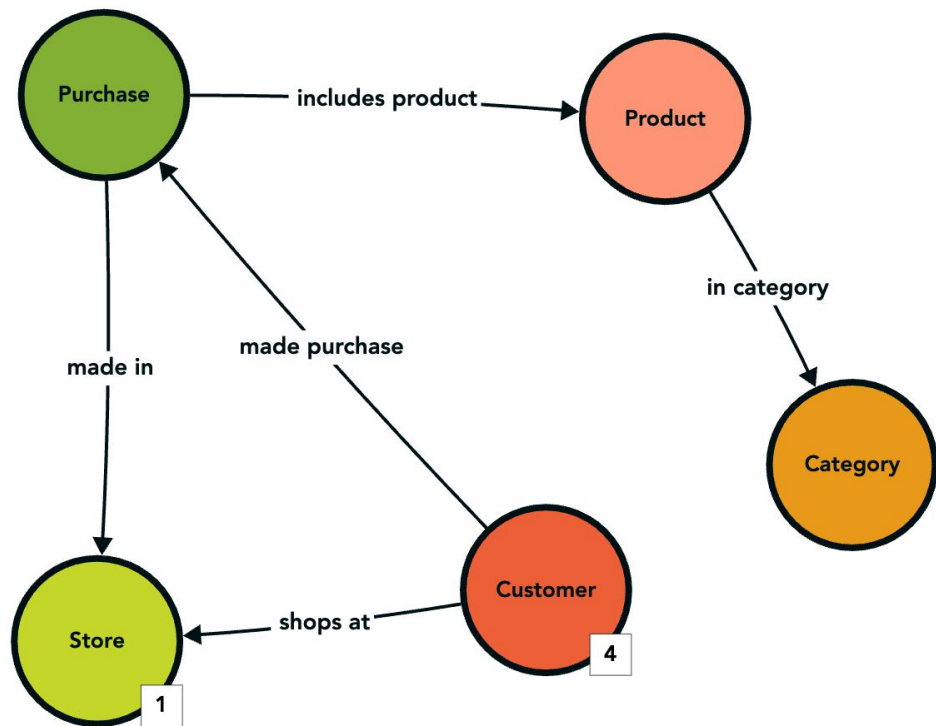
Identify the Attributes

Identify the **noun (subj)**, **verbs and** **attributes**:

- Customer **made** a purchase
- Customer **made** purchase with a **Loyalty Account**
- Products **are** in a category
- Categories **have** a parent category
- Customer **shops** in a Store **[physical store]**
- Customer **shops** at a Store **[online]**
- Customer **is** Gender **[female]**
- Purchase **made** with a **Credit Card****[visa, mastercard, amex]**
- Purchase **made** in Year **[2022]**
- *In store shoppers* are **Customers** who have an *in store purchase*

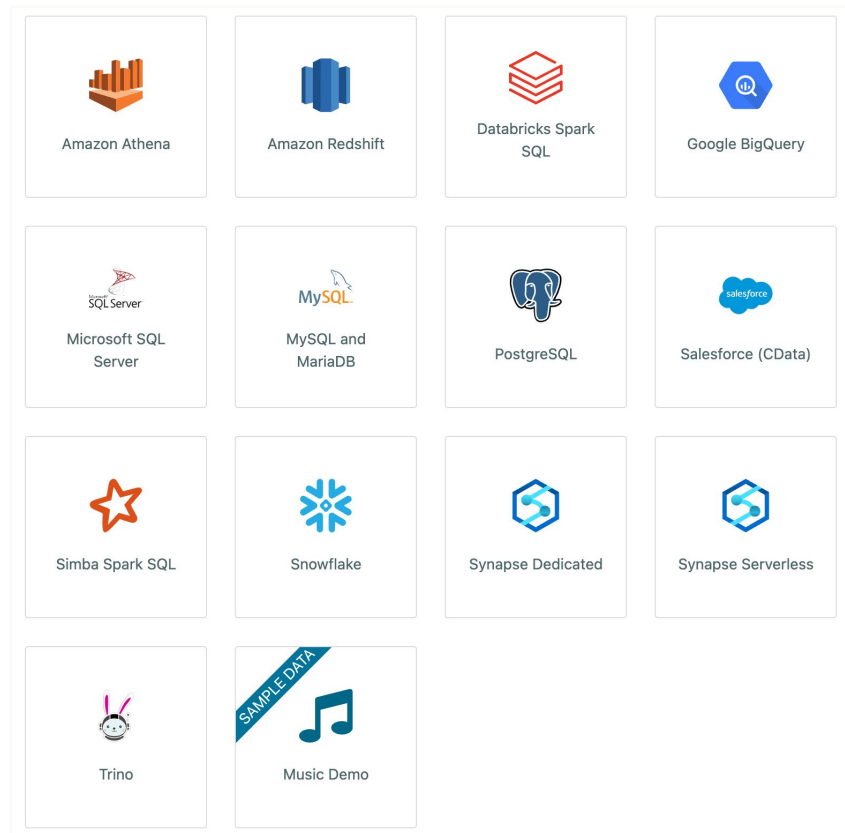
Sketch the Data Model

- Create Classes for the nouns
- Connect the Classes with the verbs (relationships)
- Create a list of attributes for each Class



Identify Data Sources

- Think about what data is needed to populate these concepts
- Create a list of these sources
- Determine if this data should be virtualized or materialized or some combination of the two



Data Modeling Terminology



Stardog is an Enterprise Knowledge
Graph based on RDF.

RDF Triple & Graph

An **RDF triple** is made up of a subject, predicate and object.

An **RDF graph** is a set of triples.

Example



RDF (Studio)



Designer



Also called



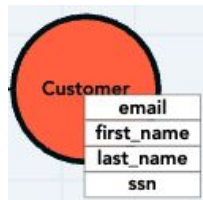
Note: In Designer, some of the naming conventions differ from the traditional RDF terms.

Triples

Modeling Terminology



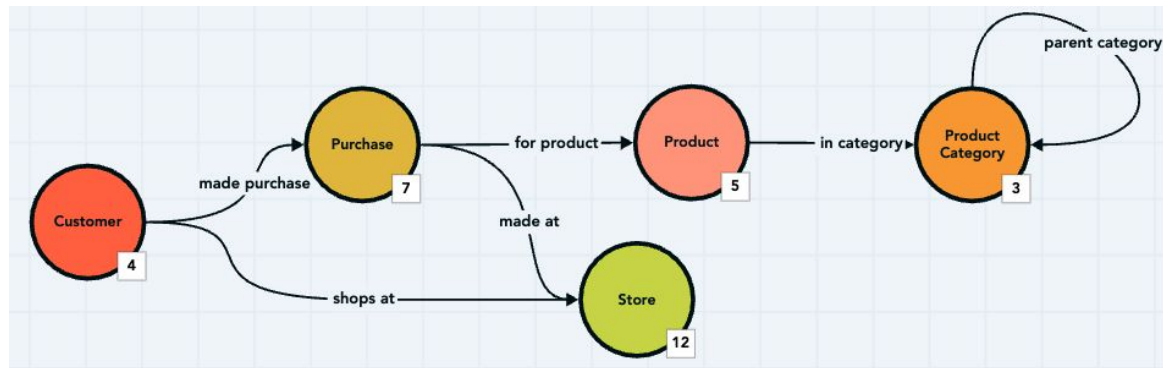
Class is a type of thing.
A class can be made up
of a set of individuals
which are called
instances.



Class with 3 attributes.
Attributes describe
properties of class
instances.

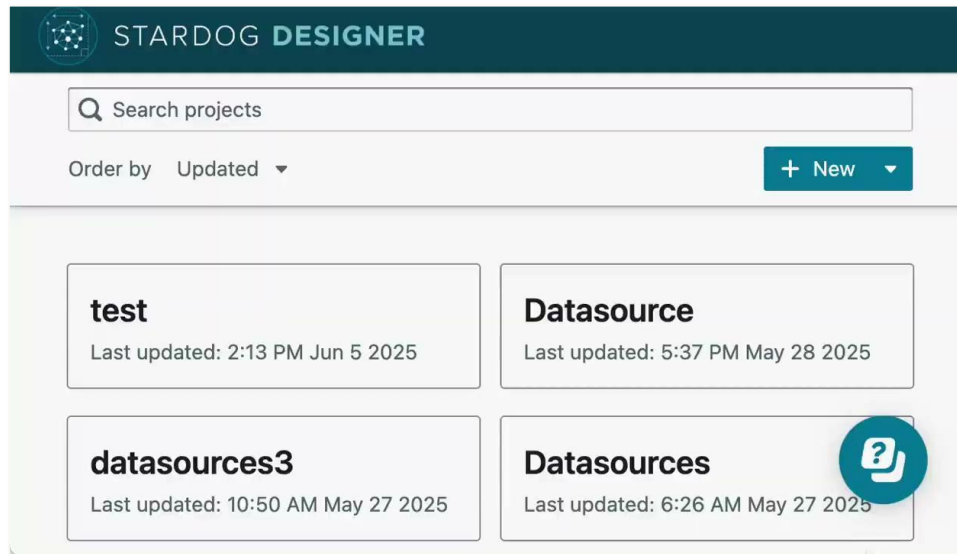


Relationships describe
an edge between 2
objects (classes or
individuals).

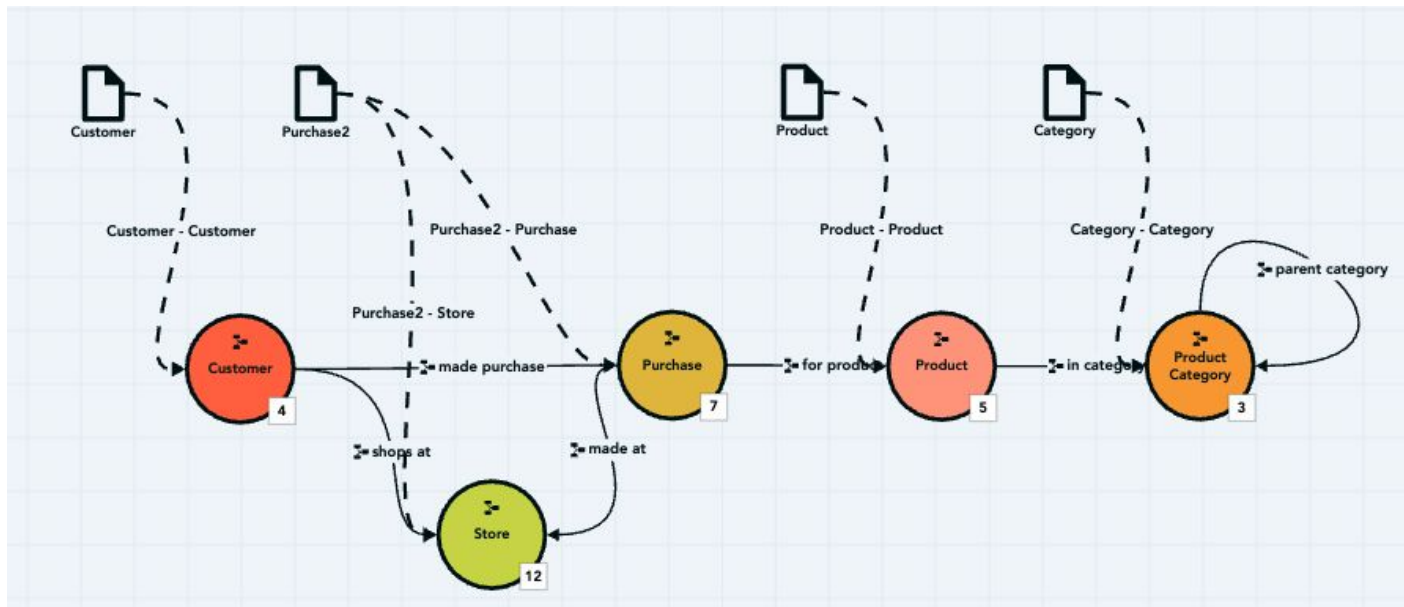
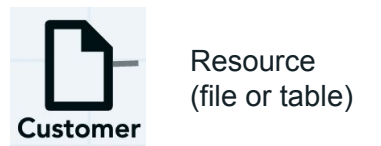


Saving Projects

- Projects can now be promoted by users, saving them to cloud storage or, if on-premise, to the account.
- Each project is specific to its account and endpoint.
- For local storage, projects will be saved within the user's browser.
- To share a project, export it as a .zip file or commit it to a source control system.



Designer Canvas Legend



Stardog Designer Model User Interface

The screenshot displays the Stardog Designer Model User Interface. The interface is divided into several sections:

- Left Sidebar:** Contains a search bar, a "Questions" section with 0 items, and a "Resources" section with 0 items. A callout points to the "Resources" section, stating "Specify data to be mapped".
- Top Bar:** Includes a "Data Model" tab and a "Mapping" tab. A callout points to the "Mapping" tab, stating "Toggle between Model overview and Mapping pages". There is also a "Publish" button. A callout points to the "Publish" button, stating "Publish project to a Stardog endpoint".
- Model Overview:** The main workspace shows a grid with a "Model: Designer Training" header. Callouts point to various icons in the toolbar:
 - "Import data models from Stardog endpoints" (Import icon)
 - "Redraw layout" (Redraw icon)
 - "Create a class" (Class icon)
 - "Create a relationship" (Relationship icon)
 - "Delete concept" (Delete icon)
- Right Sidebar:** Contains a "Rules" section. A callout points to the "Rules" section, stating "Manage rules". There is also a "Flip between Tabular / Graph view" button. A callout points to this button, stating "Flip between Tabular / Graph view".
- Bottom Bar:** Includes a "Voicebox" icon. A callout points to the "Voicebox" icon, stating "Open Voicebox".

Tabular View of the Data Model

The tabular view shows Classes, Relationships and Attributes defined in your Designer project.



Classes Relationships Attributes

Columns

Class Name	Attributes	KG Training...
Category	3	●
Customer	4	●
Product	5	●
Purchases	7	●

Classes Relationships Attributes

Columns

Relationship Name	Sources	Targets	KG Training...
for product	Purchases	Product	●
in category	Product	Category	●
made purchase	Customer	Purchases	●
parent category	Category	Category	●

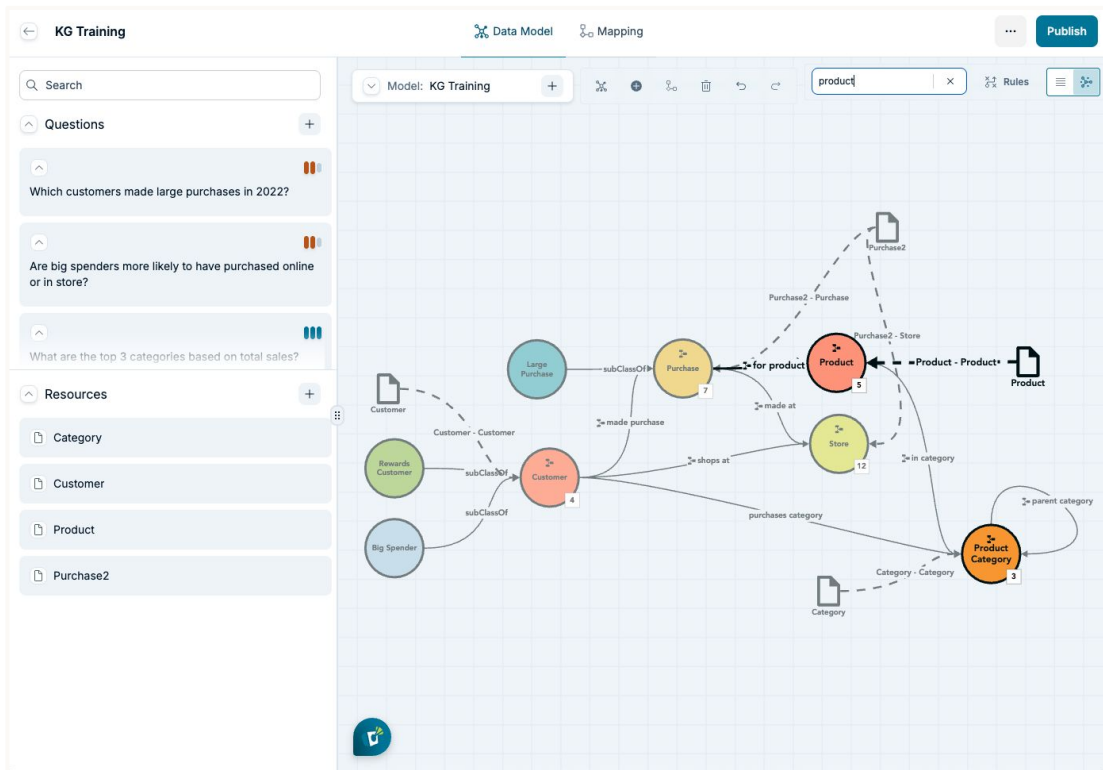
Classes Relationships Attributes

Columns

Attribute Name	Data Types	Associated Clas...	KG Training...
card	string	Purchases	●
cid	integer	Purchases	●
date	date	Purchases	●
dept	integer	Product	●
dept_name	string	Category	●
desc	string	Product	●
description	string	Category	●
email	string	Customer	●
first_name	string	Customer	●
id	integer		●
last_name	string	Customer	●
msrp	float	Product	●
parent	string	Category	●
pid	integer	Purchases	●
price	float	Purchases	●
product	string	Product	●
quantity	integer	Purchases	●
reward_account	string	Purchases	●
ssn	string	Customer	●

Search Across the Canvas

The tabular view shows Classes, Relationships and Attributes defined in your Designer project.



Edit Classes

- Edit label and description
- Customize IRI
- Manage parent and child classes
- Add relationships
- Manage attributes
- Set Geospatial visualization parameters
- Delete class

Class Detail

Basic Hierarchy Relationships Attributes Geospatial

Class Name

Purchase

Description

Delete

Edit Relationships

- Edit label and description
- Customize IRI
- Manage source and target classes
- Delete relationship

Relationship Detail

Relationship Name

made purchase

Description

Source

Customer

X

Target

Purchase

X

Source	Target
Customer	Purchase

Delete

Edit Attributes

- Edit label and description
- Customize IRI
- Edit Data Type
- Manage class associations
- Manage sensitive property group member
- Delete attribute

< Attribute Detail

Attribute Name

email

Description

Data Type

string

Associations

Customer

Sensitive Property Groups

+ Create

Filter

Name

Other attributes

Delete

The Store Use Case

Purchase

id	pid	date	time	quantity	price	cid	card	reward_accd	Store_Type	StoreID	Store_URL	Store_Address	Store_City	Store_Zip	Store_State
bad78090-e8	176	8/8/19	7:45 AM	11	9722.24	206	fcc08543-ef9e-42a9-b6ac		Online	1	https://store.example.com				
9ebf4c10-95	220	4/19/19	12:35 PM	19	11941.69	18	44f8a099-74c3f53240-13		Online	1	https://store.example.com				
20578c3c-1b	237	7/2/21	4:48 AM	15	11397.45	257	ea40b1a1-7437483d4d-57		Online	1	https://store.example.com				
f69b0158-58	522	8/16/12	10:56 PM	22	16439.94	342	ca34a80a-e360555aef-3a		Online	1	https://store.example.com				
93b06e51-bc	183	5/20/15	5:55 PM	25	18927.75	278	762a0179-9e81eb3dd1-55		Physical Stor	7		6145 Main St	San Jose	95101	CA
beed44b7-fd	160	5/15/15	6:38 PM	8	2177.84	220	5878519f-f679-47fb-9702-		Physical Stor	2		1242 Main St	New York	10001	NY

Store

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

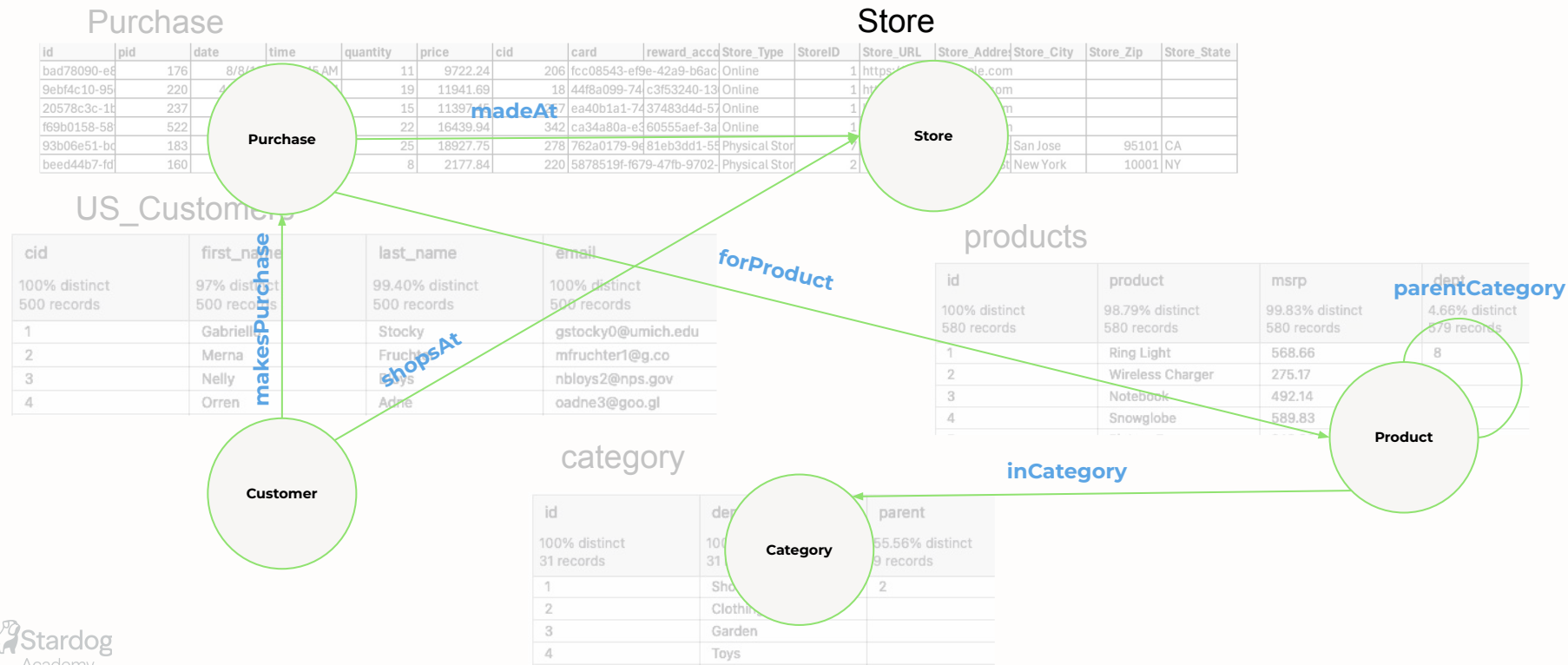
products

id	product	msrp	dept
100% distinct 580 records	98.79% distinct 580 records	99.83% distinct 580 records	4.66% distinct 579 records
1	Ring Light	568.66	8
2	Wireless Charger	275.17	8
3	Notebook	492.14	23
4	Snowglobe	589.83	21

category

id	dept_name	parent
100% distinct 31 records	100% distinct 31 records	55.56% distinct 9 records
1	Shoes	2
2	Clothing	
3	Garden	
4	Toys	

The Store Use Case



Exercise



- Open the Exercise Guide
- Complete Exercises 1, 2, & 3

02

Project Resources & Mapping

Where Mapping Sits

Mapping happens at the junction between the 2 worlds of RDB and Graph

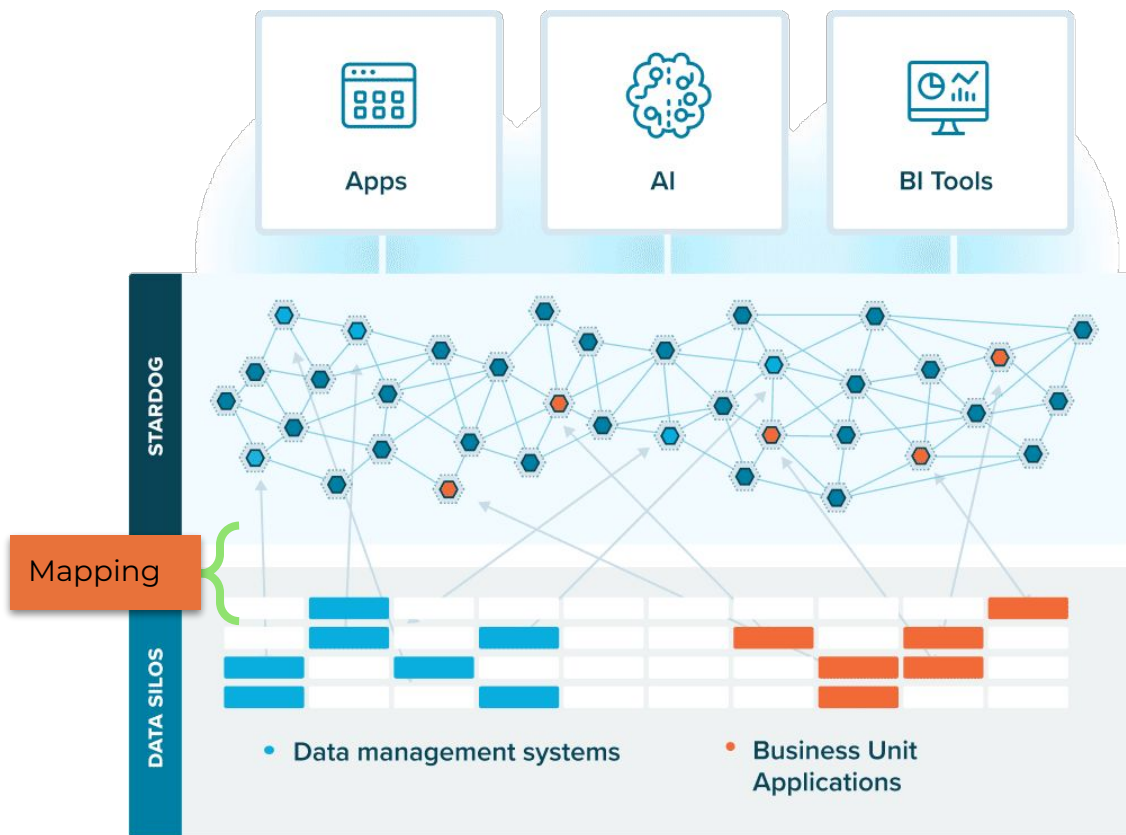
In your data you have

- Files and Tables
- Column
- Joins

That will become or be “mapped to”

- Classes
- Attributes
- Relationships

In the graph world

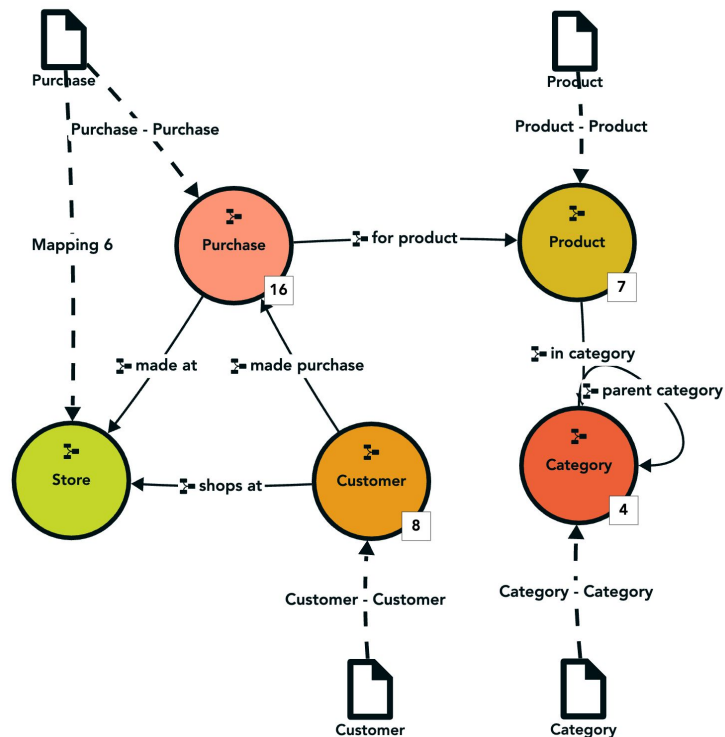


Mapping Project Resources

Designer facilitates mapping new data sources to Stardog.

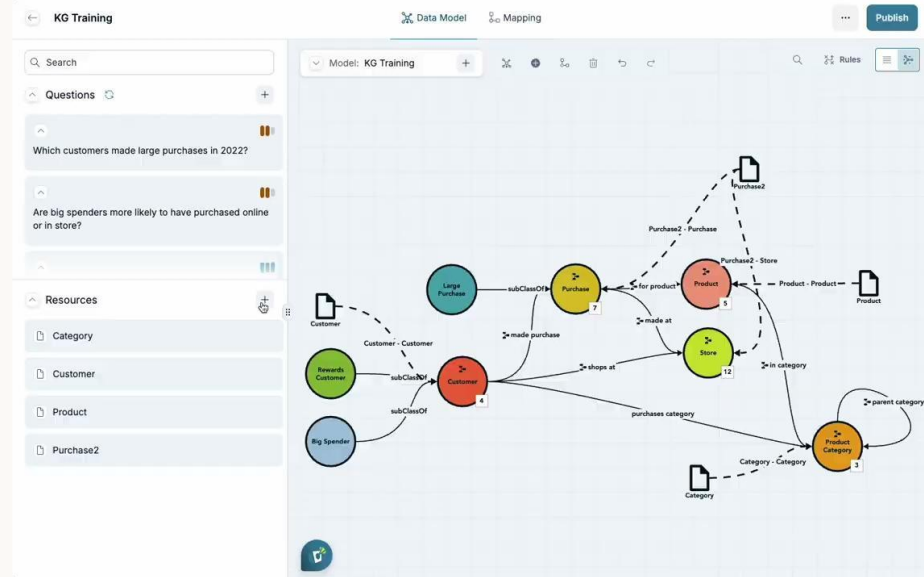
Supported data sources include:

- CSVs
- JDBC Data Connections



CSVs

- Rename resource and column headers
- See summary statistics and data preview
- Customize separator, quote, and escape characters
- Upload multiple CSV resources at the same time



Edit CSV Resources

- Edit resource name
- Edit resource property
- Mappings are preserved when possible
- Replace CSV with a new file

File Upload Preview

Source
products.csv
Replace

Resource name
products

Resource Properties

Separator character
Quote character
Escape character
Has header
Skip empty values

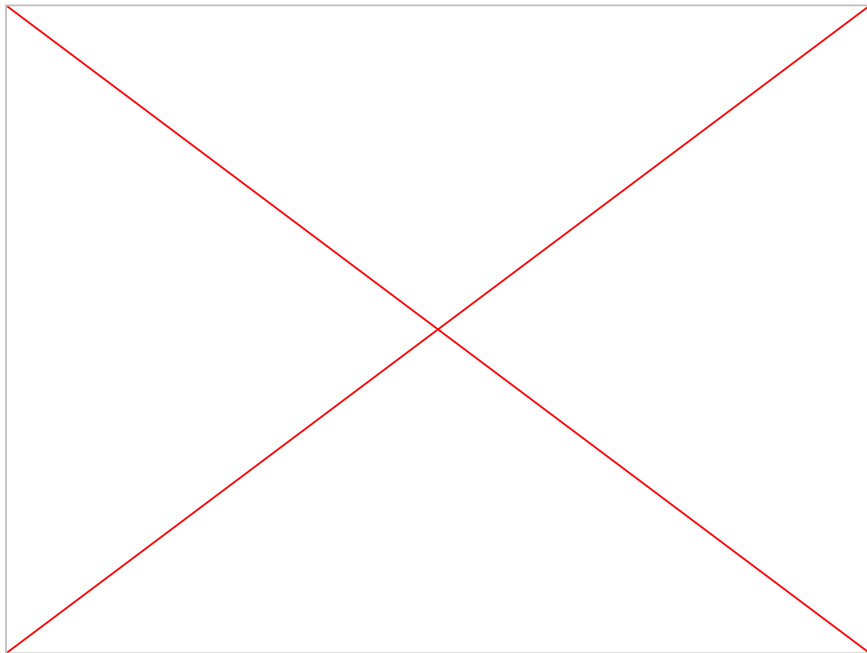
Preview 25 samples

productID	productName	quantityPerUnit	unitPrice	discontinued	categoryID
100% distinct 77 records	100% distinct 77 records	90.91% distinct 77 records	80.52% distinct 77 records	2.60% distinct 77 records	10.39% distinct 77 records
1	Chai	10 boxes x 20 bags	18	0	1
2	Chang	24 - 12 oz bottles	19	0	1
3	Aniseed Syrup	12 - 550 ml bottles	10	0	2
4	Chef Anton's Cajun S...	48 - 6 oz jars	22	0	2
5	Chef Anton's Gumbo...	36 boxes	21.35	1	2
6	Grandma's Boysenbe...	12 - 8 oz jars	25	0	2
7	Uncle Bob's Organic ...	12 - 1 lb. pkgs.	30	0	7
8	Northwoods Cranber...	12 - 12 oz jars	40	0	2
9	Mishi Kobe Niku	18 - 500 g pkgs.	97	1	6
10	Ikura	12 - 200 ml jars	31	0	8
11	Queso Cabrales	1 kg pkg.	21	0	4
12	Queso Manchego La ...	10 - 500 g pkgs.	38	0	4
13	Konbu	2 kg box	6	0	8
14	Tofu	40 - 100 g pkgs.	23.25	0	7
15	Genen Shoyu	24 - 250 ml bottles	15.5	0	2
16	Pavlova	32 - 500 g boxes	17.45	0	3

Cancel
Save

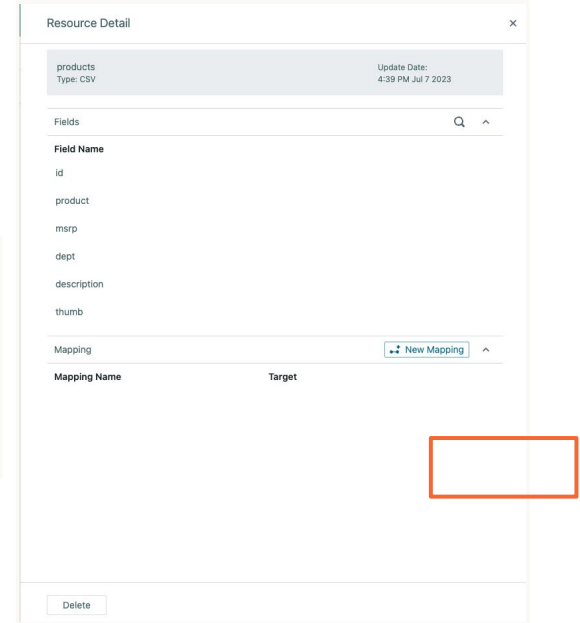
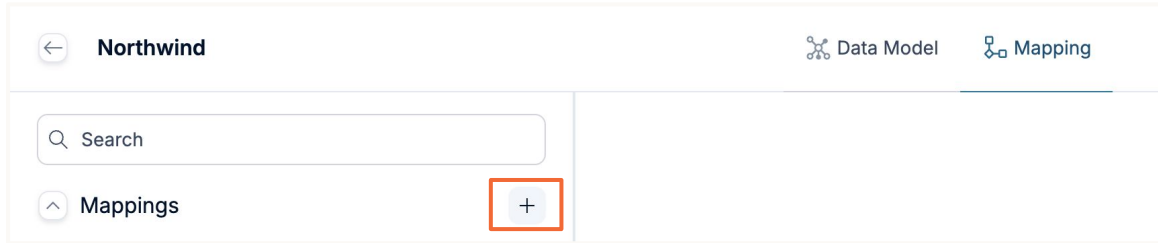
Virtual Graphs via JDBC Data Connections

- Select database tables or select data with a custom query
- Rename resource and column headers for VGs and CSV
- Includes Data preview



Add mapping

- **New Mapping** button on resource details
- **New Mapping** button on Mapping view



Stardog Designer Mapping User Interface

The screenshot displays the Stardog Designer Mapping User Interface. On the left is a navigation tree showing a project structure with categories like 'Category', 'Customer', 'Product', and 'Purchase2'. A callout bubble points to this tree, stating: 'Navigation tree of project resource, class, and mappings'. At the top, a button labeled 'Create new mappings' is highlighted with a callout. The main area is split into two tabs: 'Data Model' and 'Mapping'. The 'Mapping' tab is active, showing a configuration for a 'Category' mapping. It includes fields for 'Primary Identifier' (set to 'id'), 'Label' (set to 'dept_name'), and 'Relationships'. The 'Relationships' section shows a mapping from 'parent category' to 'parent category' with a 'Target Identifier' of 'parent'. A callout bubble points to the 'Mapping' tab, stating: 'Toggle between Model overview and Mapping pages'. At the bottom, the 'Attributes' section lists 'dept_name', 'description', and 'parent' as string attributes.

Create new mappings

Navigation tree of project resource, class, and mappings

Toggle between Model overview and Mapping pages

Category - C...
Product Cate...
Edit Attributes

Category
Type: CSV

id x

Preview: 1

Label i
dept_name

Relationships

in category

Source Class

parent category

Target Class
Product Category

Target Identifier
parent x

Preview: 2

parent category

Source Class

purchases category

Source Class

Attributes

dept_name string dept_name

description string description

parent string parent

Exercise



- Complete Exercise 4 - Upload Your Data
 - Upload and map data
- Complete Exercise 5 - Edit Classes

03

Relationships

The Store Use Case

Purchase

id	pid	te	time	quantity	price	cid	card	reward_account	Store_Type	Store_URL	Store_Address	Store_City	Store_Zip	Store_State
bad78090-e88e-	176	8/8/19	7:45 AM	11	9722.24	206	fcc08543-ef9e-42a9-b6ac-5b913df		Online	https://store0.example.com				
9ebf4c10-95de-4	220	4/19/19	12:35 PM	19	11941.69	18	44f8a099-74c3f53240-136c-4e0b-		Online	https://store1.example.com				
20578c3c-1b31-	237	7/2/21	4:48 AM	15	11397.45	257	ea40b1a1-7437483d4d-5758-468b-		Online	https://store2.example.com				
f69b0158-58f4-4	522	8/16/12	10:56 PM	22	16439.94	342	ca34a80a-e360555aef-3a70-4216-		Online	https://store3.example.com				
93b06e51-bc44-	183	5/20/15	5:55 PM	25	18927.75	278	762a0179-9e81eb3dd1-5583-4785-		Physical Store		6145 Main St	San Jose	95101	CA

US_Customers

cid	first_name	last_name	email
100% distinct 500 records	97% distinct 500 records	99.40% distinct 500 records	100% distinct 500 records
1	Gabriello	Stocky	gstocky0@umich.edu
2	Merna	Fruchter	mfruchter1@g.co
3	Nelly	Bloys	nbloys2@nps.gov
4	Orren	Adne	oadne3@goo.gl

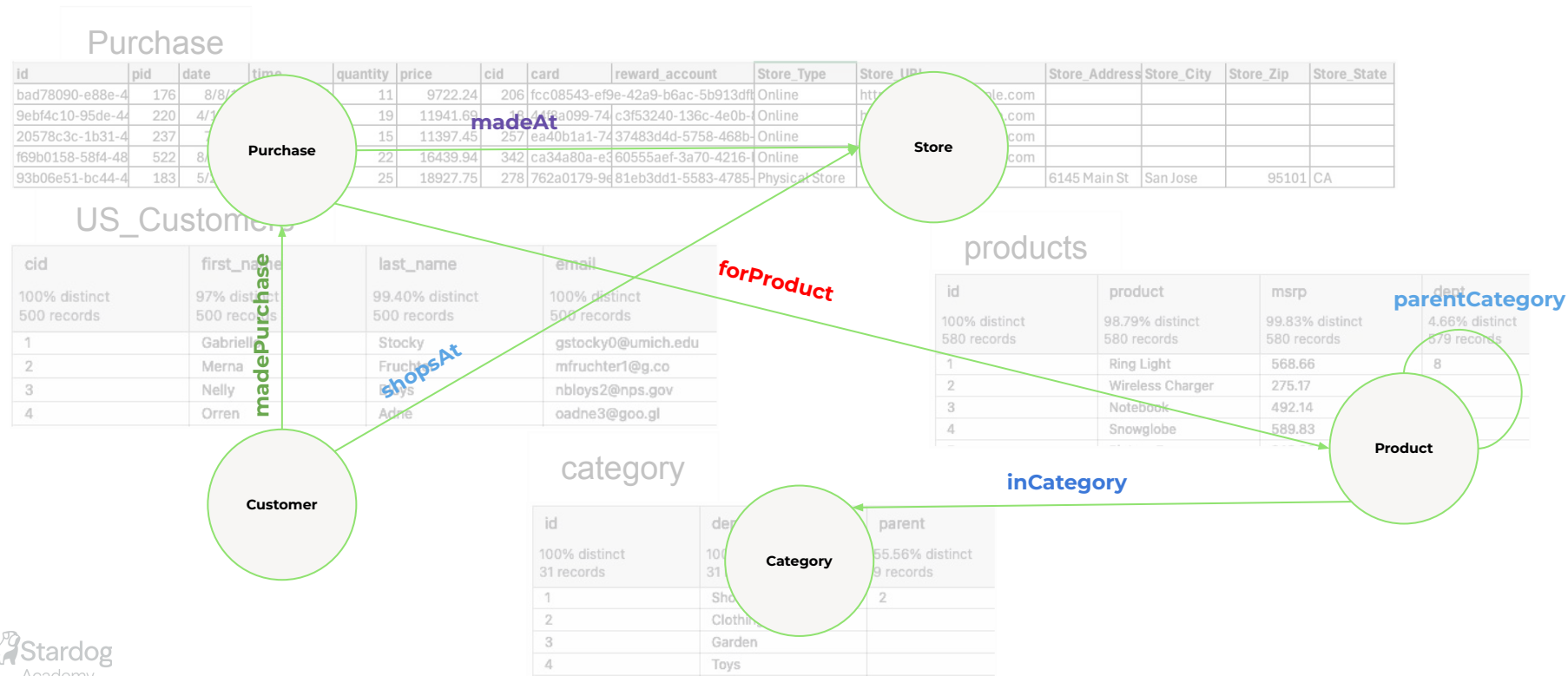
products

id	product	msrp	dept
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category

id	dept_name	parent
100% distinct 31 records	100% distinct 31 records	55.56% distinct 9 records
1	Shoes	2
2	Clothing	
3	Garden	
4	Toys	

The Store Use Case



Map the Relationship

Map project resources to relationships.

- Mapping Toggle
- Fill in the target class & target identifier under relationships
- Version 11.1+: map to a static string

The screenshot shows the 'Mapping' tab in the Stardog interface for a project named 'KG Training'. The left sidebar contains a tree view of the project's ontology, with 'Purchase2' selected. The main area is divided into sections for 'Purchase2' (Type: CSV), 'Primary Identifier', 'Label', 'Relationships', and 'Attributes'.

Purchase2 (Type: CSV) [Suggest Mapping](#)

Primary Identifier
Id [x](#)
Preview: bad78090-e88e-40f5-ad75-5329445a739e

Label
☒ Id [x](#)

Relationships

Relationship	Target Class	Target Identifier	Preview
for product	Product	pid pid x	176
made at Target Class Store mapped in Purchase2 - Store	Store	1 1 x	urn:KG_Training:Store: 1 reset
made purchase	Customer	cld cld x	206

Attributes

Attribute	Type	Value	Preview
card	string	card card x	...
cld	integer	cld cld x	...
date	date	date date x	...

Exercise



- Complete Exercise 6 - Create and Map Relationships
- Review

04

Publish the Model

Publish the Model

Database

- Select or create a database
- Download project files
 - Same files as the export option



Publish Project

Database Properties

Configure Data Models

Configure Mappings

Database

KG_Training

Enable Voicebox

Publish questions as spotlight questions

Download zipped project when publishing

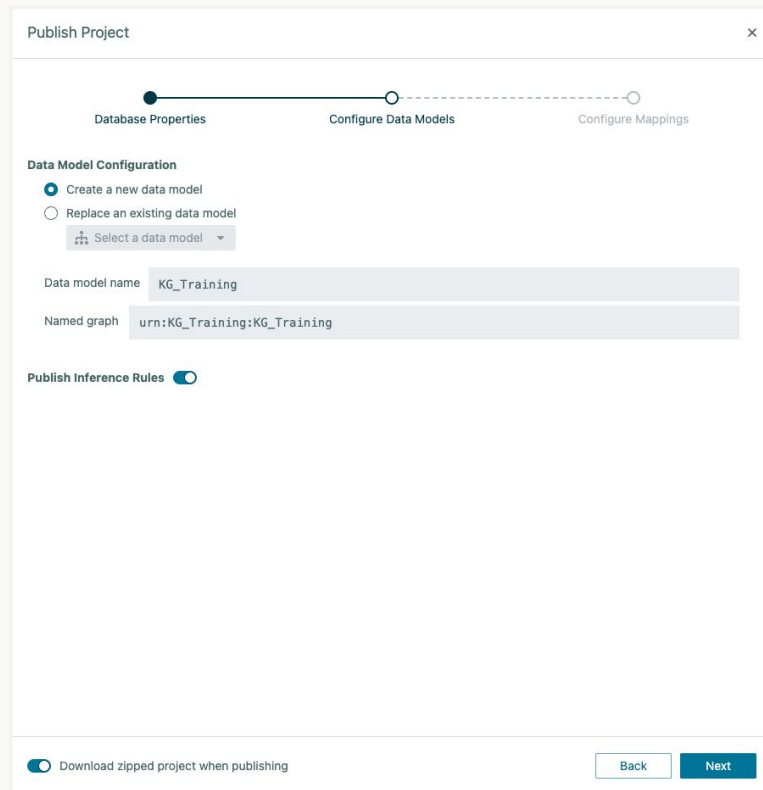
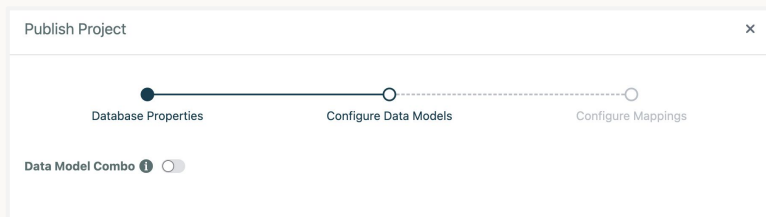
Back

Next

Publish the Model

Configure Data Models

- Create or Replace the data model
- Edit data model name
- Edit Named Graph
- (Optional) Publish Data Model Combos



Configure Mappings

- Choose how to publish resources
 - Create new
 - Replace
 - Append
- Customize adv resource details
- Publish

Publish Project ×

Database Properties ● ● Configure Data Models ○ Configure Mappings

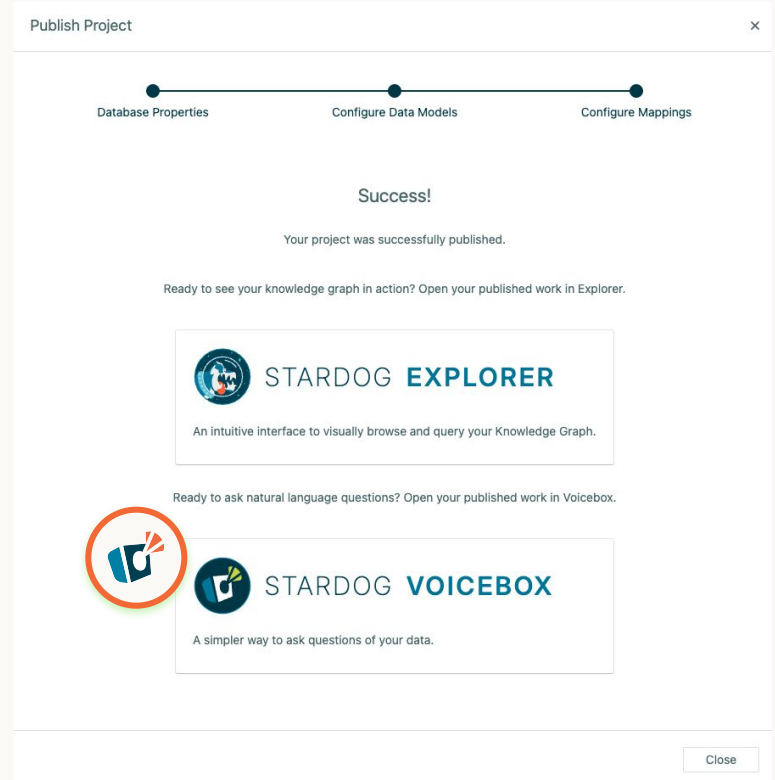
Publish	Resource	Mappings	Configuration
☒	Customer	Customer - Customer	☒ Create new ☐ Replace ☐ Append ^
Named graph urn:KG_Training:Customer			
☒	Product	Product - Product	☒ Create new ☐ Replace ☐ Append v
☒	Category	Category - Category	☒ Create new ☐ Replace ☐ Append v
☒	Purchase2	Purchase2 - Purchase Purchase2 - Store	☒ Create new ☐ Replace ☐ Append v

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Back Publish

Success!

- Exported project files downloaded locally if option was checked
- Launch Applications
 - Stardog Explorer
 - Stardog Voicebox

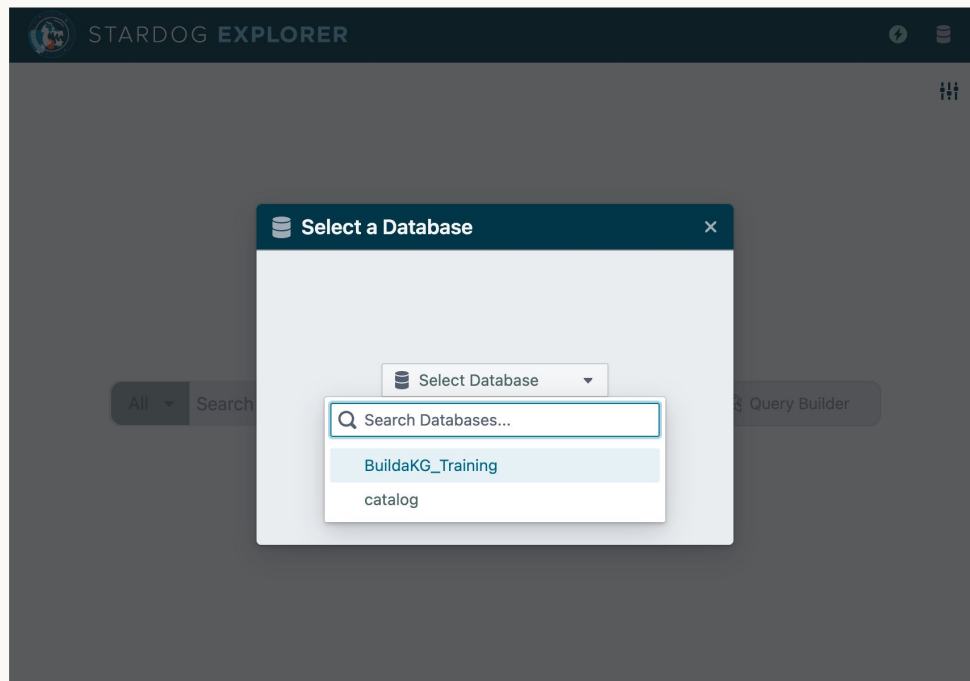


Project Export

- Data Model
 - In Turtle (.ttl) format
- Project Resource mappings
 - Mapping file in Stardog Mapping Syntax (SMS) format
 - Properties file
 - (For CSV resources) Original CSV
- Stardog Designer Project definition
 - With extension `.stardogdesigner`

Validate Model in Explorer

- Go to <https://cloud.stardog.com>
- Open Explorer
- Select database
- Select visualize

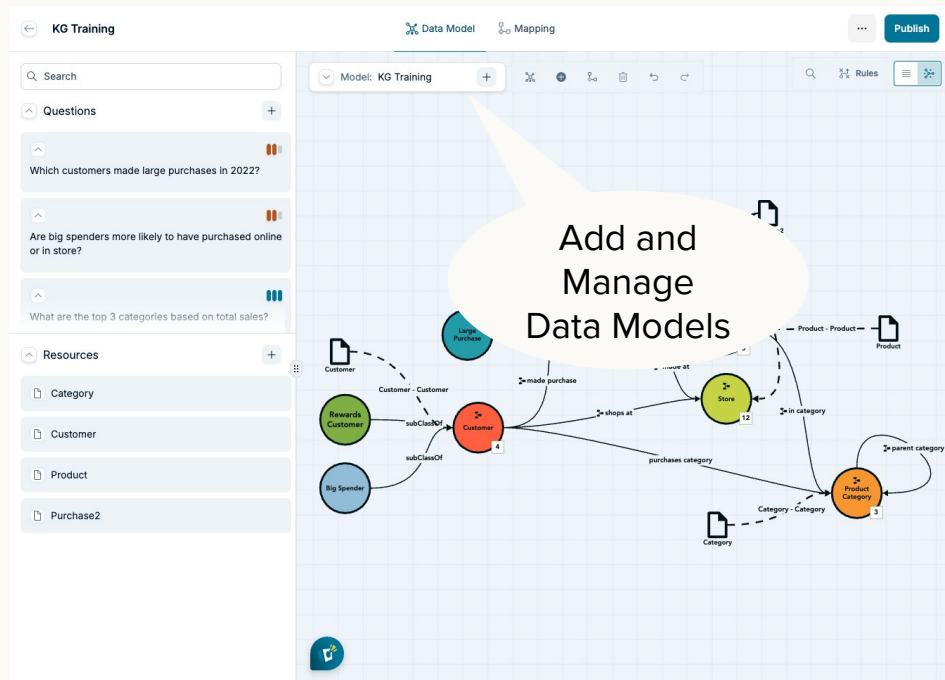


Exercise

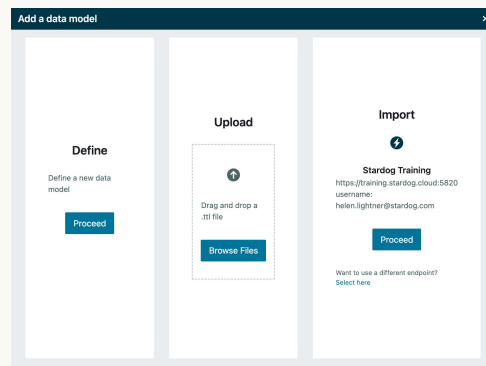


- Complete Exercise 7 - Publish your project
- Review

Managing Multiple Models in Designer

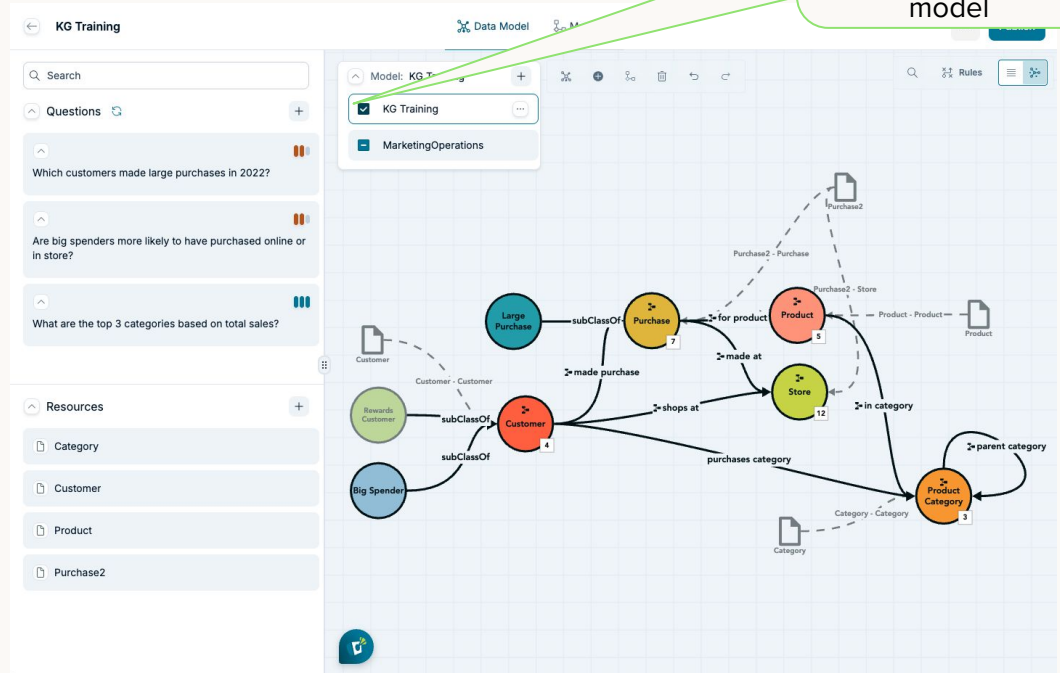
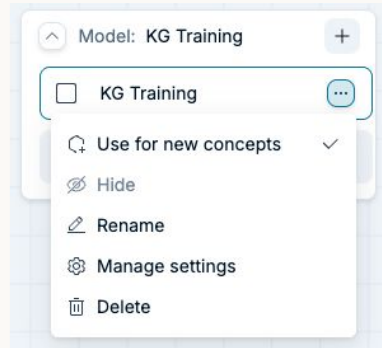


- Add a data model by building a new one or importing one from another database or a .ttl



Working with Multiple Data Models

- Add multiple data models to the canvas
- ... data model options



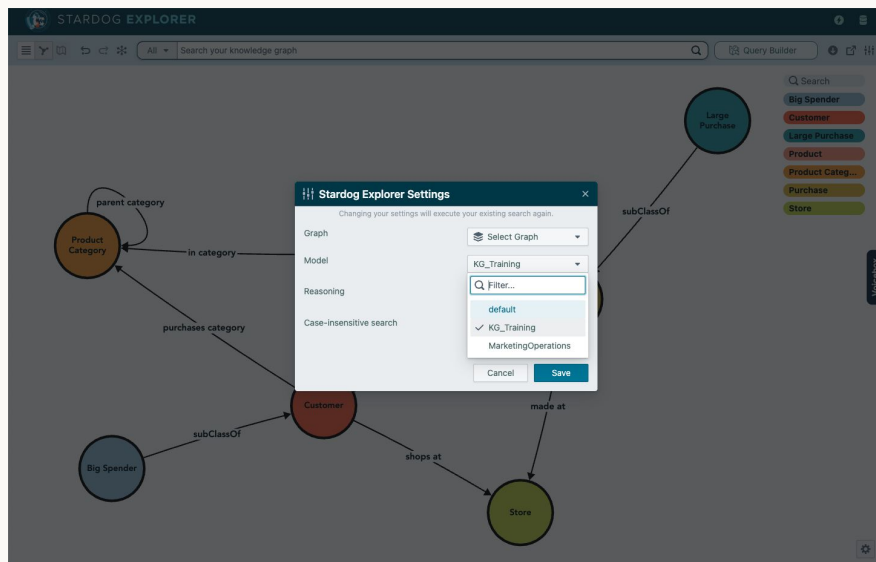
Publishing Multiple Data Models (Designer)

- Publish multiple data models
- Both models can be accessed in Settings under Model

The 'Publish Project' dialog shows a progress bar with three steps: Database Properties, Configure Data Models (current), and Configure Mappings. Under 'Data Model Selection', 'Multiple data models' is selected. Under 'Data Model Configuration', two models are listed: 'KG_Training' and 'MarketingOperations'. For 'KG_Training', 'Create a new data model' is selected, and 'KG_Training' is chosen from the dropdown. The 'Data model name' is 'KG_Training' and the 'Named graph' is 'urn:KG_Training:KG_Training'. For 'MarketingOperations', 'Create a new data model' is selected, and 'Select a data model' is chosen from the dropdown. The 'Data model name' is 'MarketingOperations' and the 'Named graph' is 'urn:KG_Training:MarketingOperations'.

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A smaller version of the 'Publish Project' dialog, showing the same progress bar and configuration options as the larger one.



Publishing Multiple Data Models (Designer)

New combo model can be accessed in the Explorer Settings or can be added to Designer as a new data model

- Publish multiple data models
- Combo data model: links 1 or more data models using a single identifier
- Create a new or replace an existing data model
- Select / Deselect data models to include

The 'Publish Project' dialog box is shown with the 'Configure Data Models' tab selected. It features a progress bar at the top with three steps: 'Database Properties', 'Configure Data Models' (current), and 'Configure Mappings'. Under 'Data Model Selection', the 'Multiple data models' radio button is selected. The 'Data Model Configuration' section has two expandable sections: 'KG Training' and 'MarketingOperations'. In 'KG Training', the 'Replace an existing data model' radio button is selected, and 'KG_Training' is chosen from the dropdown. The 'Data model name' is 'KG_Training' and the 'Named graph' is 'urn:KG_Training:KG_Training'. In 'MarketingOperations', the 'Create a new data model' radio button is selected, and 'Select a data model' is chosen from the dropdown. The 'Data model name' is 'MarketingOperations' and the 'Named graph' is 'urn:KG_Training:MarketingOperations'. At the bottom, there is a checkbox for 'Download zipped project when publishing' and 'Back' and 'Next' buttons.

The 'Publish Project' dialog box is shown with the 'Include Data Models' tab selected. It features a progress bar at the top with three steps: 'Database Properties', 'Configure Data Models' (current), and 'Configure Mappings'. Under 'Data Model Combo', the 'Data Model Combo' checkbox is checked. The 'Create a new data model' radio button is selected, and 'KG_Training_data_model_combo' is chosen from the dropdown. The 'Data model name' is 'KG_Training_data_model_combo'. Below this, the 'Included data model named graphs' section shows a table with three rows, each with a checked checkbox, a 'Data model named graph', and a 'Data Model(s)'.

	Data model named graph	Data Model(s)
☒	urn:KG_Training:KG_Training	default, KG_Training
☒	urn:KG_Training:MarketingOperations	MarketingOperations

The 'Stardog Explorer Settings' dialog box is shown. It has a title bar with the Stardog logo and 'Stardog Explorer Settings'. Below the title bar, a message says 'Changing your settings will execute your existing search again.' The settings are organized into sections: 'Graph' with a 'Select Graph' dropdown, 'Model' with a dropdown showing 'KG_Training_data...' and a search filter 'Filter...', 'Reasoning' with a 'default' dropdown, and 'Case-insensitive search' with a checkbox. A 'Marketin' dropdown is also visible at the bottom.

Publishing Multiple Data Models (Designer)

- Consolidate to one data model
 - Combines selected data models into one new model
 - A new model is created or an existing one is replaced

Publish Project

Database Properties Configure Data Models Configure Mappings

Data Model Selection

Select how to publish data models

☐ Multiple data models ☒ Consolidate to a single data model

Data Model Configuration

☒ Create a new data model
☐ Replace an existing data model

☒ kg_training

Data model name: kg_training_model

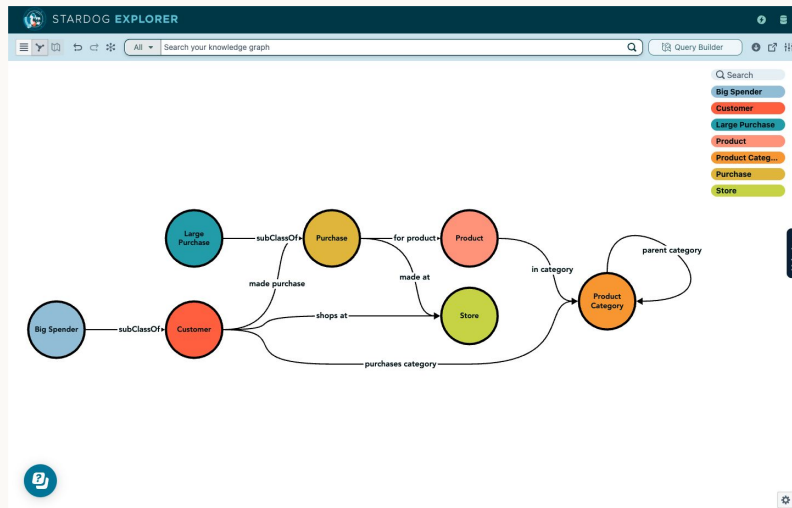
Named graph: urn:model:kg_training_data_model

Publish Inference Rules: ☒

☒ Download zipped project when publishing

Back Next

- Access new model in Explorer Settings



Tabular View

- View and search for Classes, Relationships and Attributes in each in each data model

The screenshot displays the Stardog Tabular View interface. On the left, there is a sidebar with a search bar and sections for 'Questions' and 'Resources'. The 'Questions' section contains three queries. The 'Resources' section lists 'Category', 'Customer', 'Product', and 'Purchase2'. The main area shows a table with columns for 'Class Name', 'Attributes', and two data models: 'KG Training' and 'Marketing...'. The table lists various classes and their associated attribute counts and status in each model.

Class Name	Attributes	KG Training	Marketing...
Big Spender	4	●	○
Customer	4	●	●
Large Purchase	7	●	○
Product	5	●	○
Product Category	3	●	○
Purchase	7	●	○
Rewards Customer	4	○	●
Store	12	●	○

05

Inference Rules

What is the power of inferencing?

Differentiates Graph from Knowledge Graph

- Make the implicit explicit
- Enables a Semantic Layer
- Formally define your business logic, **no-code**
 - Use your model to infer new edges and classes
 - Rules can be saved to one or more data models
- Inferences are computed at **query time**
 - Enables Data Fabric with Virtualization

Inference Rules

Identify statements that will be answered by applying business logic:

- In store shoppers are **Customers** who have an in store purchase

Rule 1: A **Customer** that makesPurchase Purchase

Purchase at a **physical store** can be classified as a Customer **In Store Shopper**

Building Rules

- View existing rules
- Build rules using a Rule Builder or a Text Editor
- Save a rule to one or more data models

The screenshot shows the 'KG Training' interface. On the left, there's a sidebar with 'Questions' and 'Resources'. The main area is divided into two panels: 'Data Model' and 'Mapping'. The 'Data Model' panel shows a graph with nodes like 'Customer', 'Big Spender', 'Rewards Customer', and 'Large Purchase', connected by relationships like 'subClassOf' and 'made purchase'. The 'Mapping' panel shows a list of 'Inference Rules' with a 'Create Rule' button.

The screenshot shows the 'Rule Detail' dialog box. It has two tabs: 'Rule Builder' and 'Text Editor'. The 'Rule Builder' tab is active, showing a rule named 'Loyalty Customer'. The rule is defined by the conditions: 'Customer' made purchase 'Purchase' and 'Purchase' has 'reward_account' exists. The rule then infers 'Customer' as 'Rewards Customer'. The dialog also has 'Delete', 'Cancel', and 'Save' buttons.

Building Rules – Rule Builder

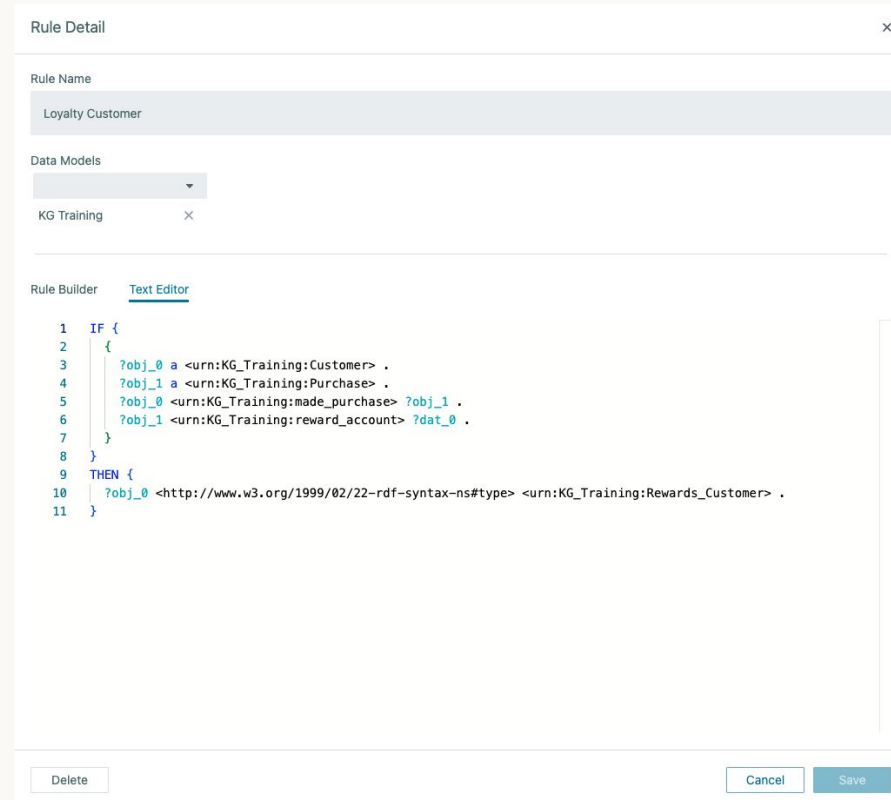
- **IF** clause specifies the conditions that are to be met. Follows the same structure as the Query Builder
- **THEN** clauses allow you
 - to classify instances as members of a class
 - to assert relationships between instances in classes
 - to assert attributes in instances of a class

The screenshot shows the 'Rule Detail' panel of a 'Rule Builder' interface. It includes a 'Rule Name' field with a 'New Rule' button, a 'Data Models' dropdown menu, and a 'KG Training' toggle. Below these are tabs for 'Rule Builder' (selected) and 'Text Editor'. The 'Rule Builder' tab contains two sections: 'When the following conditions are met:' and 'Then infer:'. Each section has a circular icon with a plus sign, a 'Select class...' dropdown menu, and a close button (X).

Tip: If you save a change, it will save the changes for the window that was open when saved

Building Rules – Text Editor

- The Text Editor uses the Stardog Rules Syntax (SRS)
- Rules include an IF and Then clause
- Syntax Highlighting
- SRS is based on W3C SPARQL Grammar

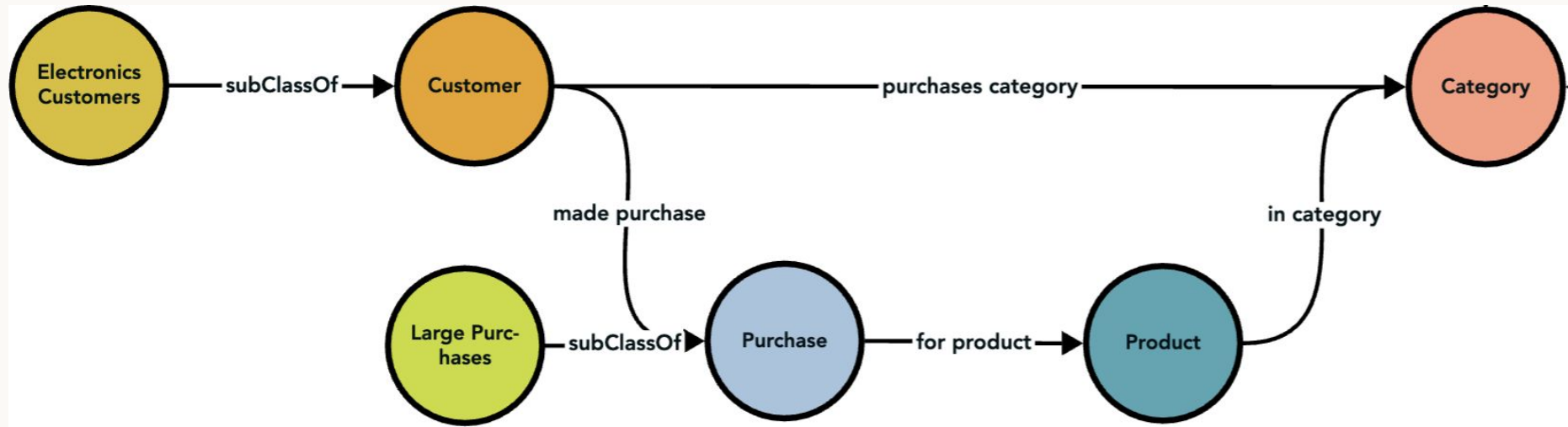


The screenshot displays the 'Rule Detail' window in Stardog. The 'Rule Name' field contains 'Loyalty Customer'. Under 'Data Models', 'KG Training' is selected. The 'Text Editor' tab is active, showing a SPARQL query with syntax highlighting and line numbers:

```
1 IF {  
2 {  
3   ?obj_0 a <urn:KG_Training:Customer> .  
4   ?obj_1 a <urn:KG_Training:Purchase> .  
5   ?obj_0 <urn:KG_Training:made_purchase> ?obj_1 .  
6   ?obj_1 <urn:KG_Training:reward_account> ?dat_0 .  
7 }  
8 }  
9 THEN {  
10  ?obj_0 <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <urn:KG_Training:Rewards_Customer> .  
11 }
```

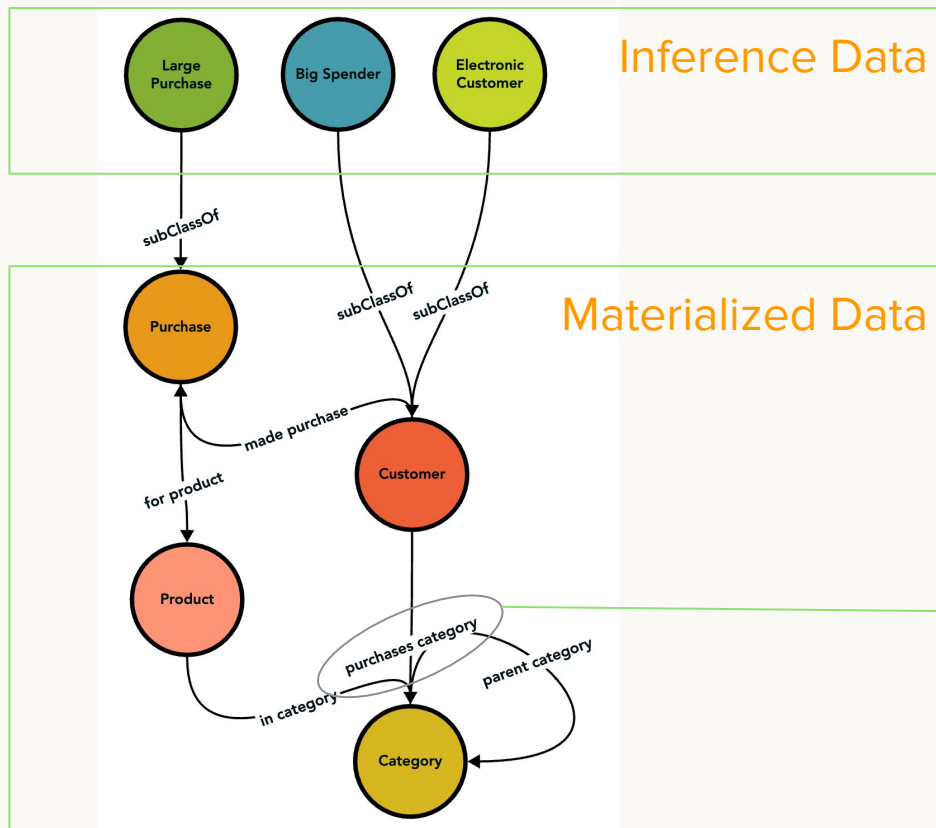
At the bottom, there are 'Delete', 'Cancel', and 'Save' buttons.

Inference Examples....



- "Electronics Customers" and "Large Purchases" are inferred classes
- "purchases category" is an inferred relationship

Store Data Layers



Exercise



- Complete Exercise 8 - Create inference rules
- Complete Exercise 9 & 10 - Publish your project and view it in Explorer

Exercise

Instructions

- Open Settings and change the model to **Model you just published** and turn reasoning on
- Build the following query using the query builder. Big Spenders that made a large orders after Jan 1, 2022. Save and run the query

Exercise

- Complete Exercise 11

IRI

Internationalized Resource Identifier (IRI) is a string used to identify nodes and edges uniquely. Using IRIs prevents name conflicts and doesn't require centralized control.

Class Detail

Class Name

Product

[DATA MODEL NAMESPACE]Product

Relationship Detail

Relationship Name

made purchase

[DATA MODEL NAMESPACE]made_purchase

Example IRI: `tag:stardog:designer:designer_training_example:data:Product`

Review

Course Review

1

Data modeling basics

2

Creating and editing the data model properties

3

Mapping project resources in Designer

4

Publishing the model and viewing in Explorer

5

Inference Rules and Reasoning

Questions?



Thank You!

To contact Stardog about training please email:
training@stardog.com

